

Pre-Integration Data Quality Assessment

Data Management Project

July 8, 2026

1 Assessment Scope

This report documents a Data Management project focused on restaurant review platforms in the Milan area. The current report is a pre-integration baseline: it assesses Google Places, Tripadvisor, and TheFork data after acquisition and before entity matching. The project can later integrate these sources to analyse cross-platform rating consistency on a unified restaurant table.

2 Data Sources

The current acquisition layer contains three raw source datasets: Google Places as geographic seed and reference source, Tripadvisor as a general-purpose review platform, and TheFork as a restaurant-specific platform. All quality metrics in this report are computed from raw acquisition outputs. No entity matching or post-integration metric is included in this baseline.

3 Pre-Integration Quality Metrics

The selected data-quality dimensions are completeness, validity/consistency, uniqueness, timeliness, and reliability. Completeness measures missing values; validity checks source-specific field formats on present values; uniqueness checks stable source identifiers; timeliness measures source refreshability from collection duration; reliability flags ratings based on sparse review evidence. All percentage component scores are oriented so that 100% is better. This is a raw pre-cleaning baseline and should be rerun after cleaning/enrichment to measure improvement with the same formulas.

3.1 Metric Definitions

Metric	Formula / definition	Reading
Completeness	$100 * \text{present relevant field values} / \text{expected relevant field values}$	Average coverage over all profiled fields.
Critical completeness	$100 * \text{present critical field values} / \text{expected critical field values}$	Coverage of fields required for matching and analysis.
Validity	$100 * \text{valid present values} / \text{present values checked by validators}$	Source-specific format checks; missingness is handled separately.
Spatial readiness	$100 * \text{records with valid latitude/longitude pair} / \text{records}$	Readiness for geospatial integration.
Timeliness	$\max(0, 100 * (1 - \text{collection_duration_hours} / \text{refresh_target_hours}))$	Refreshability against the current 48.00h target.
Reliability	$100 * \text{records with review_count} \geq \text{low_review_threshold} / \text{records}$	Share of ratings supported by enough reviews.
Uniqueness	$100 - 100 * \text{duplicate flags} / \text{records}$	Penalizes duplicate identifiers and normalized name/address keys.
Quality flags	$\text{flags_per_100} = 100 * \text{generated warnings} / \text{records}$	Record-level warnings for cleaning, not a deletion list.

3.2 Cross-Source Summary

Source	Records	Score	Complete	Critical	Validity	Spatial	Timeliness	Reliable	Flags/100
Google Places	10808	93.16%	93.79%	98.15%	100.00%	100.00%	67.23%	79.50%	27.2
Tripadvisor	7539	72.28%	84.34%	99.78%	99.99%	0.00%	43.75%	53.10%	48.0
TheFork	1344	97.63%	76.15%	98.88%	100.00%	100.00%	95.83%	88.84%	17.0

3.3 Weighted Quality Score

The weighted score favours dimensions required for integration and analysis: critical completeness, validity, spatial readiness, uniqueness, timeliness, and review-count reliability. This avoids treating sparse optional metadata as equal to identifiers, ratings, review counts, and coordinates. Average completeness is reported separately, while critical completeness is the completeness component used in the weighted score.

Source	Score	Critical	Validity	Spatial	Uniqueness	Timeliness	Reliability
Google Places	93.16%	98.15%	100.00%	100.00%	99.85%	67.23%	79.50%
Tripadvisor	72.28%	99.78%	99.99%	0.00%	99.97%	43.75%	53.10%
TheFork	97.63%	98.88%	100.00%	100.00%	100.00%	95.83%	88.84%

3.4 Pre-Integration Visual Diagnostics

The following generated visuals summarize source readiness before matching: weighted score bars, component-level diagnostics, core-field coverage, and the most frequent anomaly classes.

Weighted quality score

Source	Score	Visual scale
Google Places	93.16%	
Tripadvisor	72.28%	
TheFork	97.63%	

Score components

Source	Component	Value	Visual scale
Google Places	Critical	98.15%	
Google Places	Validity	100.00%	
Google Places	Spatial	100.00%	
Google Places	Uniqueness	99.85%	
Google Places	Timeliness	67.23%	
Google Places	Reliability	79.50%	
Tripadvisor	Critical	99.78%	
Tripadvisor	Validity	99.99%	
Tripadvisor	Spatial	0.00%	
Tripadvisor	Uniqueness	99.97%	
Tripadvisor	Timeliness	43.75%	
Tripadvisor	Reliability	53.10%	
TheFork	Critical	98.88%	
TheFork	Validity	100.00%	
TheFork	Spatial	100.00%	
TheFork	Uniqueness	100.00%	
TheFork	Timeliness	95.83%	
TheFork	Reliability	88.84%	

Core coverage heatmap

Source	ID	Name	Address	Rating	Reviews	Coords	Timestamp
Google Places	100.00%	100.00%	100.00%	93.51%	93.51%	100.00%	67.23%
Tripadvisor	100.00%	100.00%	99.12%	99.80%	100.00%	0.00%	43.75%
TheFork	100.00%	100.00%	100.00%	94.12%	98.07%	100.00%	95.83%

Colored markers indicate pre-integration readiness bands.

Anomaly profile

Source	Issue type	Count	Visual scale
Google Places	low_review_count	1515	
Google Places	missing_critical_field	1402	
Google Places	possible_duplicate_name_address	20	
Tripadvisor	low_review_count	2723	
Tripadvisor	zero_reviews	813	
Tripadvisor	missing_critical_field	81	
Tripadvisor	possible_duplicate_name_address	4	
TheFork	low_review_count	117	
TheFork	missing_critical_field	105	
TheFork	zero_reviews	7	

3.5 Comparative Findings

Comparison	Finding
Largest source	Google Places with 10808 records.
Highest weighted quality score	TheFork (97.63%).
Lowest weighted quality score	Tripadvisor (72.28%).
Highest completeness	Google Places (93.79%).
Weakest completeness	TheFork (76.15%).
Highest critical completeness	Tripadvisor (99.78%).
Coordinate-ready sources	Google Places, TheFork; Tripadvisor still requires latitude/longitude enrichment.
Tripadvisor raw limitation	Lower raw score is expected: no coordinates, no record-level timestamps, and many zero/low-review records before cleaning.
Most sparse review evidence	Tripadvisor (36.12%).
Highest quality-flag density	Tripadvisor (48.04 flags per 100 records).
Weakest source refreshability	Tripadvisor (43.75%).

3.6 Quality Improvement Actions

Priority	Scope	Action	Expected effect
1	Tripadvisor	Geocode addresses or match records to Google Places before spatial joins.	Raises spatial readiness and enables automated integration-error checks.
2	Google Places	Document and reduce collection duration, or add incremental refresh.	Improves source refreshability when frequent updates are required.
2	Tripadvisor	Document and reduce collection duration, or add incremental refresh.	Improves source refreshability when frequent updates are required.
2	Tripadvisor	Persist record-level scrape/acquisition timestamps and parse them in QA.	Keeps traceability separate from the refreshability score.
3	Google Places	Use review-count weighted ratings for cross-platform comparisons.	Reduces the impact of sparse ratings on consistency analysis.
3	TheFork	Fix or document the empty contact fields from the latest scrape.	Prevents website, social-link, phone, or email fields from being used as match evidence.
3	Tripadvisor	Use review-count weighted ratings for cross-platform comparisons.	Reduces the impact of sparse ratings on consistency analysis.
4	Google Places	Inspect duplicate normalized name/address keys before integration.	Prevents false entity matches and duplicated restaurants.
4	Tripadvisor	Inspect duplicate normalized name/address keys before integration.	Prevents false entity matches and duplicated restaurants.

4 Source-Specific Assessment

4.1 Google Places

Google Places is treated as the broad geographic seed and reference source. Its coordinate coverage makes it the most useful starting point for spatial integration, while weak optional fields mostly affect enrichment depth.

Key metrics	Metric		Value
	Records assessed		10808
	Overall quality score		93.16%
	Average field completeness		93.79%
	Critical-field completeness		98.15%
	Validity score		100.00%
	Uniqueness score		99.85%
	Timeliness score		67.23%
	Collection duration		15.73 h
	Timestamp coverage		100.00%
	Reliability score		79.50%
	Format values checked		192603 / 192603 valid
	Native rating average		4.26
	Comparable rating average 0–5		4.26
	Valid review counts		10107 (93.51%)
	Low-review records		1515 (14.99%)
	Valid coordinate pairs		10808 (100.00%)
	Latest source timestamp		2026-05-25T10:15:47.447928+00:00
	Data age in days		15.19
	Quality flags		2937 (27.2 per 100 records)

Weakest field coverage	Field	Present	Missing	Coverage
	website	5753	5055	53.23%
	price	8423	2385	77.93%
	phone_number	9027	1781	83.52%
	opening_hours	9427	1381	87.22%
	reviews	10106	702	93.50%
	rating	10107	701	93.51%
	review_count	10107	701	93.51%

Most frequent quality flags	Issue type		Count
	low_review_count		1515
	missing_critical_field		1402
	possible_duplicate_name_address		20

Interpretation

- The weighted source quality score is 93.16%; the score is driven by integration-critical fields rather than optional metadata.
- The weakest fields are website (53.23%), price (77.93%), phone_number (83.52%); downstream logic should avoid treating them as mandatory matching keys.

- Format validity checks 192603 present values; 0 fail the expected source-specific format.
- The sparse-review threshold flags 1515 records, so rating comparisons should account for review-count reliability.
- Timeliness score is 67.23%, based on collection duration 15.73 h against a 48.00 h refresh target; timestamp coverage is 100.00%.
- The source has 2937 generated quality flags, equal to 27.17 flags per 100 records.
- Because valid coordinates are complete, Google can anchor the first geospatial join and help detect out-of-area records in other sources.

4.2 Tripadvisor

Tripadvisor contributes broad review and category information, but the raw file does not yet contain latitude and longitude or record-level timestamps. Its lower raw score is expected and should be improved through the planned cleaning and geocoding enrichment before reliable geospatial integration.

Key metrics	Metric	Value
	Records assessed	7539
	Overall quality score	72.28%
	Average field completeness	84.34%
	Critical-field completeness	99.78%
	Validity score	99.99%
	Uniqueness score	99.97%
	Timeliness score	43.75%
	Collection duration	27.00 h
	Timestamp coverage	0.00%
	Reliability score	53.10%
	Format values checked	82654 / 82655 valid
	Native rating average	3.91
	Comparable rating average 0–5	3.91
	Valid review counts	7539 (100.00%)
	Low-review records	2723 (36.12%)
	Valid coordinate pairs	0 (0.00%)
	Latest source timestamp	n/a
	Data age in days	n/a
	Quality flags	3622 (48.0 per 100 records)

Weakest field coverage	Field	Present	Missing	Coverage
	email	3535	4004	46.89%
	opening_hours	5096	2443	67.60%
	price_range	5101	2438	67.66%
	cuisine_type	5859	1680	77.72%
	photo_count	5949	1590	78.91%
	website	6023	1516	79.89%
	reviews	6683	856	88.65%

Most frequent quality flags	Issue type	Count
	low_review_count	2723
	zero_reviews	813
	missing_critical_field	81
	possible_duplicate_name_address	4
	invalid_field_format	1

Interpretation

- The weighted source quality score is 72.28%; the score is driven by integration-critical fields rather than optional metadata.
- The weakest fields are email (46.89%), opening_hours (67.60%), price_range (67.66%); downstream logic should avoid treating them as mandatory matching keys.
- Format validity checks 82655 present values; 1 fail the expected source-specific format.
- The sparse-review threshold flags 2723 records, so rating comparisons should account for review-count reliability.
- Timeliness score is 43.75%, based on collection duration 27.00 h against a 48.00 h refresh target; timestamp coverage is 0.00%.
- The source has 3622 generated quality flags, equal to 48.04 flags per 100 records.
- The 0% coordinate validity is expected at this stage and should be resolved by the latitude/longitude enrichment task before integration.
- The missing record-level timestamps do not force timeliness to 0 anymore; Tripadvisor refreshability is estimated from the scraper runtime instead.

4.3 TheFork

TheFork is the smallest source but is highly restaurant-specific and already contains near-complete coordinates. Its ratings use a native 0-10 scale, so comparison with Google and Tripadvisor requires scale normalization.

		Metric	Value
Key metrics		Records assessed	1344
		Overall quality score	97.63%
		Average field completeness	76.15%
		Critical-field completeness	98.88%
		Validity score	100.00%
		Uniqueness score	100.00%
		Timeliness score	95.83%
		Collection duration	2.00 h
		Timestamp coverage	100.00%
		Reliability score	88.84%
		Format values checked	21493 / 21493 valid
		Native rating average	8.86
		Comparable rating average 0–5	4.43
		Valid review counts	1318 (98.07%)
		Low-review records	117 (8.88%)
		Valid coordinate pairs	1344 (100.00%)
		Latest source timestamp	2026-06-06T20:35:04+00:00
		Data age in days	2.76
		Quality flags	229 (17.0 per 100 records)

		Field	Present	Missing	Coverage
Weakest field coverage		email	0	1344	0.00%
		phone_number	0	1344	0.00%
		social_links	0	1344	0.00%
		website	0	1344	0.00%
		discount	650	694	48.36%
		opening_hours	855	489	63.62%
		rating	1265	79	94.12%

		Issue type	Count
Most frequent quality flags		low_review_count	117
		missing_critical_field	105
		zero_reviews	7

Interpretation

- The weighted source quality score is 97.63%; the score is driven by integration-critical fields rather than optional metadata.
- The weakest fields are email (0.00%), phone_number (0.00%), social_links (0.00%); downstream logic should avoid treating them as mandatory matching keys.
- Format validity checks 21493 present values; 0 fail the expected source-specific format.
- The sparse-review threshold flags 117 records, so rating comparisons should account for review-count reliability.
- Timeliness score is 95.83%, based on collection duration 2.00 h against a 48.00 h refresh target; timestamp coverage is 100.00%.

- The source has 229 generated quality flags, equal to 17.04 flags per 100 records.
- TheFork is suitable for geospatial matching now, but its missing contact fields (website, social links, phone, and email) limit its usefulness for contact-based validation.

5 Complete Field Coverage by Source

5.1 Google Places

Field	Present	Missing	Coverage
place_id	10808	0	100.00%
name	10808	0	100.00%
address	10808	0	100.00%
city	10808	0	100.00%
latitude	10808	0	100.00%
longitude	10808	0	100.00%
rating	10107	701	93.51%
review_count	10107	701	93.51%
types	10808	0	100.00%
primary_type	10784	24	99.78%
details	10808	0	100.00%
phone_number	9027	1781	83.52%
website	5753	5055	53.23%
opening_hours	9427	1381	87.22%
reviews	10106	702	93.50%
price	8423	2385	77.93%
business_status	10789	19	99.82%
details_fetched_at	10808	0	100.00%
seed_collected_at	10808	0	100.00%

5.2 Tripadvisor

Field	Present	Missing	Coverage
source_url	7539	0	100.00%
restaurant_name	7539	0	100.00%
address	7473	66	99.12%
rating	7524	15	99.80%
review_count	7539	0	100.00%
cuisine_type	5859	1680	77.72%
price_range	5101	2438	67.66%
photo_count	5949	1590	78.91%
website	6023	1516	79.89%
phone_number	6795	744	90.13%
email	3535	4004	46.89%
opening_hours	5096	2443	67.60%
reviews	6683	856	88.65%

5.3 TheFork

Field	Present	Missing	Coverage
source_id	1344	0	100.00%
restaurant_url	1344	0	100.00%

Field	Present	Missing	Coverage
restaurant_name	1344	0	100.00%
address	1344	0	100.00%
city	1344	0	100.00%
latitude	1344	0	100.00%
longitude	1344	0	100.00%
rating	1265	79	94.12%
review_count	1318	26	98.07%
cuisine_type	1323	21	98.44%
price_range	1344	0	100.00%
discount	650	694	48.36%
photo_count	1343	1	99.93%
website	0	1344	0.00%
social_links	0	1344	0.00%
phone_number	0	1344	0.00%
email	0	1344	0.00%
opening_hours	855	489	63.62%
reviews	1299	45	96.65%
detail_scraped	1344	0	100.00%
scraped_at	1344	0	100.00%

6 Next Steps

After geospatial integration and entity matching, this report should be extended with match-rate metrics, unmatched and ambiguous record counts, and cross-platform rating-disagreement metrics on the unified restaurant table. In particular, Tripadvisor latitude and longitude enrichment is required before the geospatial integration step can be evaluated fairly.

7 Reproducibility

The report is regenerated by running `report/build_report.ps1` from PowerShell. The script profiles the three raw datasets, refreshes the Markdown and LaTeX tables, and then compiles `report/main.pdf`.